(Please write your Enrollment Number)

Enrollment No. 02701022022

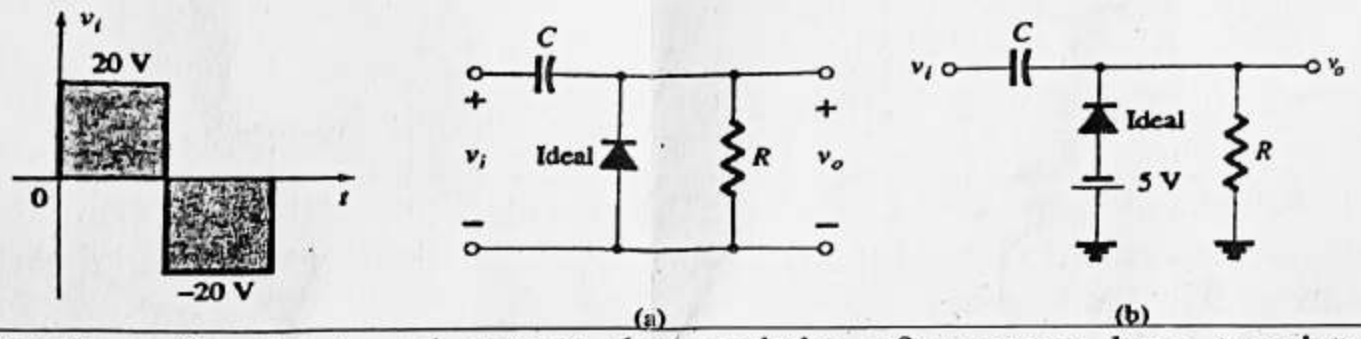
MID-TERMEXAMINATION
(CBCS)
Course Name: B.Tech. ECE, Semester: 3rd Semester
(October, 2023)

Subject Code: BEC-201 | **Subject: Analog Electronics**

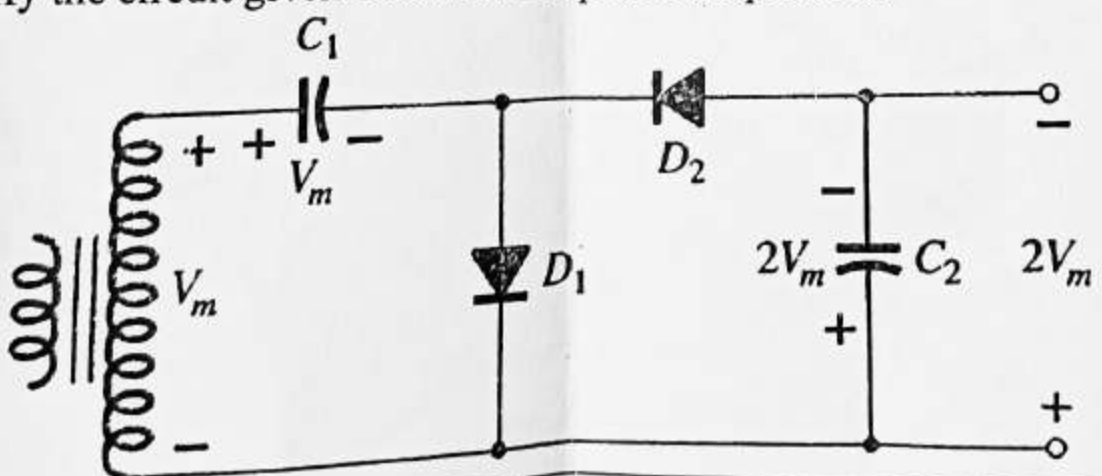
Time : 1 ½ Hours | **Maximum Marks : 30**

Note: Q. 1 is compulsory.

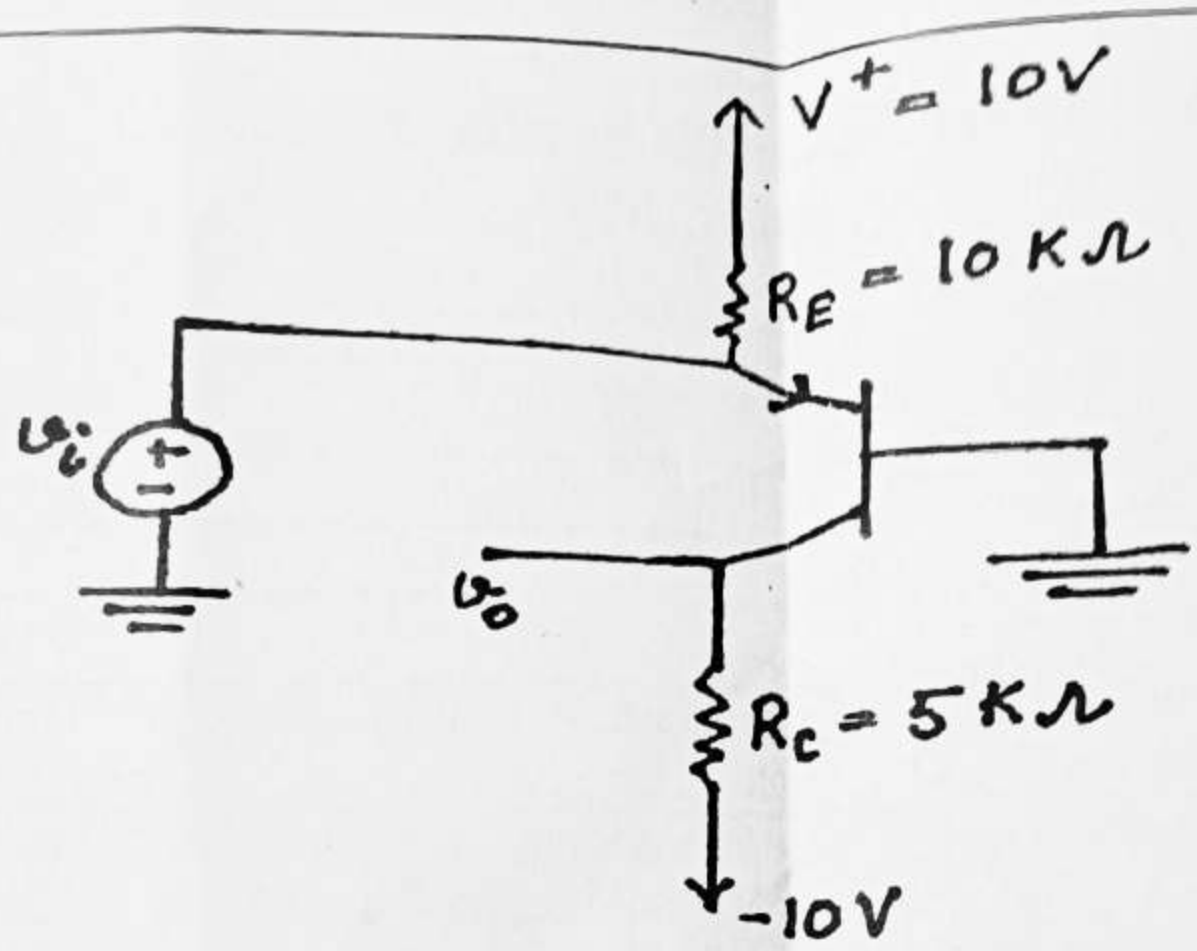
Q1			
	(a) Explain the operation of a Darlington pair. Why is it called as "Super β" configuration. Justify with required analysis.	**(2.5*4)**	
	(b) What does PIV rating mean in a rectifier circuit? Why does the PIV of the diodes in a full wave rectifier need to be 2×Vm where Vm is the peak voltage of the input AC signal.		
	(c) What is biasing? Why do we need biasing? Name few biasing techniques commonly used in BJT circuits.		
	(d) Derive the relation between small signal base resistance and small signal emitter resistance.		
Q2	**(Attempt any Two Parts) UNIT-1**	**(5,5)**	
	(a) Sketch the output voltage waveform for each network of Figure given below and explain their operation.		



	(b) Analyze the input and output characteristics of common base transistor amplifier configuration with the help of V-I waveforms. Explain different regions of operations with detailed working of CB transistor.		
	(c) Identify the circuit given below and explain its operation.		



Q3	**(Attempt any Two Parts) UNIT-2**	**(5,5)**	
	(a) Analyze the circuit given below using small signal modelling (SSM) in amplifiers & find its voltage gain.		



(b) Analyze the small signal common emitter BJT configuration and derive the following characterizing parameters for the same:-

(i) Input Resistance (R_{in})
(ii) Output Resistance (R_o)
(iii) Open Circuit Voltage Gain (A_{vo})
(iv) Closed Circuit Voltage Gain (A_v)
(v) Overall Voltage Gain (G_v)

(c) Differentiate between Cascade and Cascode amplifiers. Give one example of each with circuit diagram. Also, discuss the operation, advantages, and applications of emitter follower circuit.